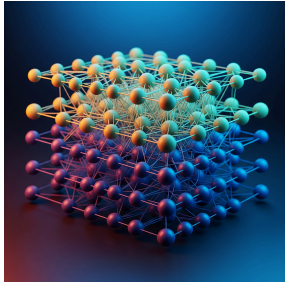


A4.3.8 Artificial Neural Networks (ANNs)



A 4.3.8 Artificial Neural Networks (ANNs)

A4.3.8 outlines Artificial Neural Networks (ANNs), which simulate interconnected "neurons" to process and learn from data. These networks enable tasks like classification, regression, and pattern recognition. The topic requires sketching a single perceptron, highlighting its input, weights, bias, activation function, and output, as well as a multi-layer perceptron (MLP) with input, hidden, and output layers to model complex patterns

Introduction

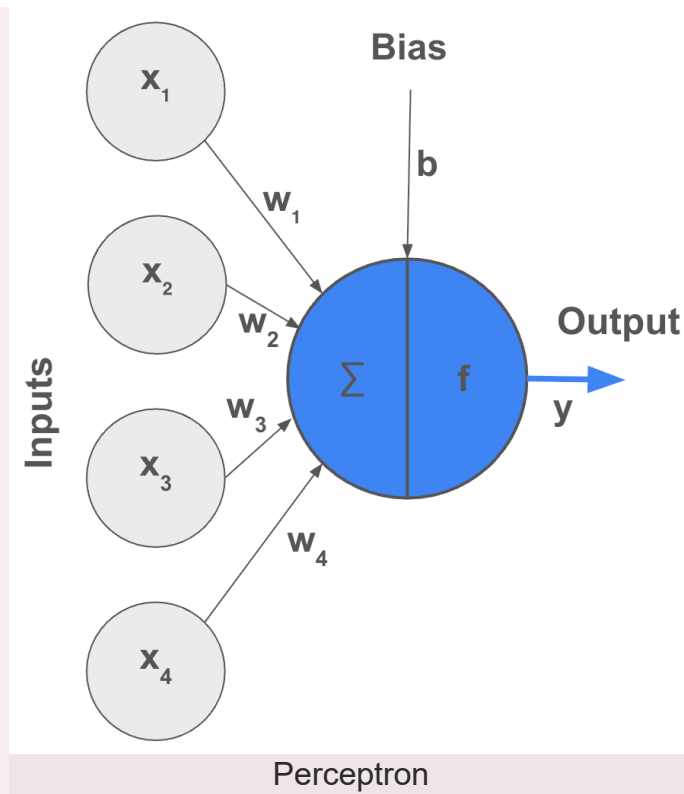
We've seen how machines can learn through different paradigms. Now, we'll explore the architecture that powers the most advanced models in machine learning, an approach inspired by the most powerful learning machine we know: the human brain. Welcome to **Artificial Neural Networks (ANNs)**.

What is an Artificial Neural Network?

An ANN is a type of machine learning model designed to simulate the way interconnected nerve cells (neurons) in the human brain process information. Instead of being programmed with explicit rules, an ANN learns to recognize patterns and make decisions by processing and learning from examples. ANNs are the foundational technology behind **Deep Learning** and are capable of performing complex tasks like classification, regression, and pattern recognition.

The Building Block: The Perceptron

The simplest form of an ANN is a single artificial neuron, known as a **Perceptron**. It's the fundamental processing unit of the network. A perceptron takes several inputs, performs a calculation, and produces a single output.



Here are its key components:

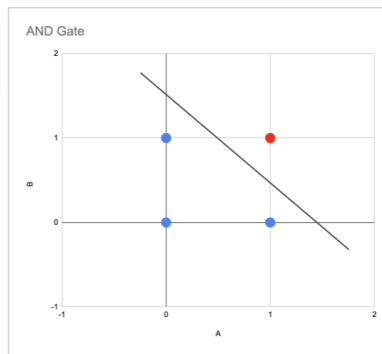
- **Inputs (x):** These are the features of your data. For example, in predicting if a student will pass an exam, the inputs could be `hours_studied` and `previous_score`.
- **Weights (w):** Each input is assigned a weight, which represents its importance. A higher weight means that input has more influence on the final decision. The network *learns* these weights during the training process.
- **Bias (b):** This is an extra, learnable parameter that acts like an adjustable threshold. It allows the neuron to shift its decision boundary, helping the model fit the data more accurately.
- **Activation Function:** After the neuron calculates the weighted sum of its inputs and adds the bias, this function makes the final "decision." It determines whether the neuron should "fire" (activate) and what its output signal should be.
- **Output (y):** The final output of the perceptron, which could be a classification (e.g., 1 for 'pass', 0 for 'fail') or a continuous value.

The Limit of a Single Perceptron

A single perceptron is a powerful building block, but it has a fundamental limitation: it can only solve problems that are **linearly separable**. This means it can only separate data into two classes if it can draw a single straight line between them.

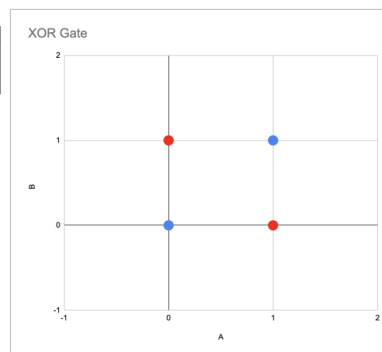
This works for simple problems like the logical AND gate, but it fails completely on problems like the XOR (Exclusive OR) gate, which is not linearly separable.

A	B	Z
0	0	0
0	1	0
1	0	0
1	1	1



Linear problem - A single straight line can easily be drawn to separate the red dot from the blue dots.

A	B	Z
0	0	0
0	1	1
1	0	1
1	1	0



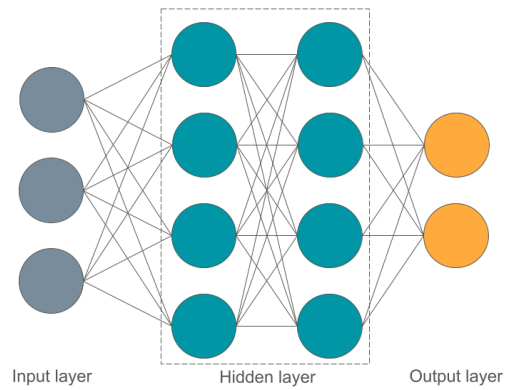
Nonlinear problem: It is impossible to draw a single straight line to separate the two colours.

The Solution: Multi-Layer Perceptrons (MLPs)

To overcome the limitation of a single perceptron, we stack them together in layers to create a **Multi-Layer Perceptron (MLP)**. This is the most common type of ANN. By adding layers, the network gains the ability to learn complex, non-linear patterns.

An MLP has three types of layers:

1. **Input Layer:** This layer receives the raw data. The number of neurons in this layer is equal to the number of features in your dataset.
2. **Hidden Layer(s):** These are the layers between the input and output. An MLP can have one or many hidden layers. It is in these layers that the network learns to find increasingly complex and abstract patterns. The presence of hidden layers is what allows an MLP to solve non-linearly separable problems like XOR.
3. **Output Layer:** This is the final layer that produces the model's prediction. The number of neurons in this layer depends on the task (e.g., one neuron for regression, multiple neurons for multi-class classification).



A Multi-Layer Perceptron

By combining many simple neurons in this layered structure, an MLP can model incredibly complex relationships in data, forming the foundation of modern deep learning.

Flashcards

Use this Quizlet to learn all the key terms in this section.



Ready to play?

Match all the terms with their definitions as fast as you can. Avoid wrong matches, they add extra time!

Start game

[View this flashcard set](#)

Choose a Study Mode 

Knowledge Check: Multiple Choice Quiz

Test your understanding of the core concepts.

1

An Artificial Neural Network is a machine learning model inspired by:

- A. ☐ The process of natural selection.
- B. ☐ The laws of physics.

C. ☐ The rules of formal logic.

D. ☒ The structure and function of the human brain.

2

In a single perceptron, what is the purpose of the "weights"?

A. ☐ To provide a baseline threshold for activation.

B. ☐ To decide if the neuron should fire.

C. ☒ To represent the importance or strength of each input signal.

D. ☐ To determine the final output category.

3

The part of a neuron that makes the final "firing" decision based on the sum of inputs is called the:

A. ☒ Activation Function

B. ☐ Weight

C. ☐ Input Layer

D. ☐ Bias

4

A single perceptron is limited because it can only solve problems that are:

A. ☐ Related to image recognition.

B. ☒ Linearly separable.

C. ☐ Unsupervised

D. ☐ Computationally expensive.

5

The classic example of a problem that a single perceptron CANNOT solve is the:

A. ☐ OR gate

B. ☐ AND gate

C. ☒ XOR gate

D. ☐ NOT gate

6

What is the key structural addition in a Multi-Layer Perceptron (MLP) that allows it to model complex patterns?

A. ☐ A higher number of weights

B. ☒ One or more hidden layers

C. ☐ A more complex activation function

D. ☐ More input neurons

7

In an MLP designed to classify images from a dataset with 784 features (pixels), the input layer should have:

A. ☐ 2 neurons

B. ☐ 1 neuron

C. ☐ 10 neurons

D. ☒ 784 neurons

8

The layers in an MLP that are responsible for learning abstract representations of the data are the:

- A. ☒ Hidden Layers
- B. ☐ Perceptron Layers
- C. ☐ Output Layers
- D. ☐ Input Layers

9

A learnable parameter in a neuron that allows it to shift its decision boundary is the:

- A. ☐ Input
- B. ☐ Output
- C. ☐ Weight
- D. ☒ Bias

10

The overall function of an ANN is to process input data to perform tasks such as:

- A. ☒ Classification, regression, and pattern recognition.
- B. ☐ Data storage.
- C. ☐ Only classification.
- D. ☐ Only regression.

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